Finite Geometry for Adaptive Contraction

A certified improvement beyond the twenty-nine reviewed SAT packets and the remaining obstacle to a smaller exponent order

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Abstract

We review twenty filed refinement packets and all nine newer archives at the pinned ProveIt snapshot, together with the public Anthropic proof supplied during this investigation. The newest comparable generic cached bound is $30.891018920366\ldots$ in the coefficient of $s/\sqrt{\log_2 s}$, with space coefficient $30.753912832055\ldots$. We give a new finite-geometric construction of adaptive contraction trees. The four parallel classes of the affine plane $\mathbb{F}_3^2$ yield a sharp local coefficient $1/\sqrt{3}$; a short Gram-matrix argument pays for every internal table construction, not only the final multiplication. Together with a second certified tree family and a fully paid recursion, the resulting normalized time coefficient is at most 30.692, with space coefficient at most 29.467, for all sufficiently large inputs. These are improvements of explicit asymptotic upper bounds. The algorithm uses subexponential space and retains the existing reductions and exact matrix-multiplication interface. It does not change the unrestricted SAT order $n \log\log n/\sqrt{\log n}$, and does not claim the requested logarithmic-order breakthrough. We also prove a scoped obstruction to a direct odd-prime replacement and extend the known binary-separator witness to unbalanced trees at the relevant arities. Exact checkers, a finite contraction prototype, provenance records, and reproducible results accompany the report. The new refinements are not Lean formalized.

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1 Outcome, comparison, and scope

There is a concrete new algorithmic ingredient in this iteration: select contraction trees from parallel classes of a finite affine plane. This gives a uniform bound over every exact mode-budget allocation, including allocations concentrated on particular pairs. Its geometric overlap properties control the maximum over all table-building operations.

The distinction between an improved exponential coefficient and a smaller exponent order is essential. Our normalized guarantee has the form

$$T(s,\mathcal{L}) \le (s+\mathcal{L})^{O(1)} 2^{(C_T+o(1))s/\sqrt{\log_2 s}}, \qquad S(s,\mathcal{L}) \le (s+\mathcal{L})^{O(1)} 2^{(C_S+o(1))s/\sqrt{\log_2 s}}.$$

Here s is a normalized Boolean-wire budget and $\mathcal{L}$ is the description length. Neither coefficient is a coefficient per SAT variable. The best polynomial-space branch remains available separately. This report does not assert a new polynomial-space coefficient.

Generic normalized branch	Time coefficient	Space coefficient
Best filed cached branch, before the nine arrivals	35.354588	34.274803
Best incoming cached branch	30.891019	30.753913
New certified cached branch	30.691338	29.466030
Best incoming polynomial-space branch	46.184846	polynomial

The first and second rows are the explicit certificates in [7, 8]. The last row is [9]. All use the generic maximum-degree promise $\lceil s^{9/10} \rceil$. Smaller numbers obtained under stronger degree promises do not compete on this same input class. An unspecified big-O constant in another algorithm cannot be ranked against these displayed coefficients. Nor are the conditional $n/\log n$ targets proved algorithms that our result dominates.

For the two explicit asymptotic upper-bound expressions being compared, reducing C_T by a positive constant reduces their ratio by an exponential factor on the $s/\sqrt{\log s}$ scale. It does not prove that an older implementation actually requires its stated upper bound, and it is not a measured SAT speedup.

1.1 Frozen repository review

The initial review pin is `73dfb5e5ab6e91bd7375e5b97ef85d7e1cc8e81d`. At that pin, the two requested directories contained twenty filed packets and nine additional ZIPs. All nine ZIPs were retrieved at immutable Git identities, checked against their Git blob hashes and ZIP integrity, and extracted into separate directories. The report sources, status files, and relevant algorithms were inspected; imported verification scripts were not treated as proofs merely because their outputs were bundled.

Commit	Relevant change
`2b925d8800cd`	First seven refinement packets placed in the project.
`3e096089911f`	OpenAI preprint vendored from the pinned upstream source.
`45233238aebf`	First-round strongest-claims summary.
`6d3adc450615`	Eight second-round packets placed.
`450252e9d86c`	Five third-round packets placed; two omitted replay scripts restored.
`b9941a7396e4`	Nine newer SAT packets added to `docs/incoming`.
`73dfb5e5ab6e`	Reviewed repository head.

These dates are October 26 in Pacific time, although their UTC dates include October 27. Immutable links and content hashes are supplied in the evidence manifest. Repository placement is distinct from mathematical verification: the ProveIt README explicitly treats the packets as externally supplied research [1].

Closing review of subsequent commits. Before delivery, commit `a7ae5ab55f96` placed the same nine archives under `Research/`, and commit `7de9d09fecd1` updated the Anthropic formalization link. The latter is the closing repository head. All 443 files actually placed in the nine new directories were compared with the reviewed archive members by Git blob identity: none changed and no unexplained new file appeared. The placement omitted 85 of the 528 archive members: duplicate source/evidence copies and one checksum ledger. All 84 omitted duplicate copies were also matched to their recorded target Git blobs. Of the 565 pre-existing files, only the README and omission ledger changed; no existing mathematical source was changed or deleted. At the closing head all twenty-nine packets are filed, and `docs/incoming` contains only its README. The mathematical comparison is unchanged; the archive contains the complete old-to-current path map [2].

1.2 Twenty filed packets

The following inventory identifies what must already be credited to the reviewed corpus. "PS" means polynomial space; "cached" allows subexponential space. Numerical entries are normalized-wire coefficients.

Packet	Contribution or limiting qualification
`cauchy_mode_improvement`	Factorial Cauchy modes; weights $1 + 3q - 2\,\mathrm{popcount}(q)$; no order change.
`eth_contraction_improvements`	Sharper Hermite precision, streamed coefficient extraction; PS 648.44.
`3sat_precision_refinement`	Leading precision $24MD/u \to 6MD/u$; faster structured summation is conditional.
`sat_density_improvement`	Uniform density-aware SAT exponent; a genuine gain at some intermediate densities.
`eth_density_refinement`	Independent density refinement; local slope-four Smith valuations.
`3sat_refinement`	Scalar parity lift and smaller incidence-degree cap; second contraction lacks a closed representation.
`3sat_asymptotic_analysis`	Restricted barriers and explicit missing degree/compression interfaces.
`eth_asymmetric_packet_balancing`	Nearly lossless asymmetric packing and thin precision gaps; PS 76.909.
`3sat_round2_refinement`	PS 84.066; actual-domain gate and independent manuscript audits.
`eth_balanced_packing`	Deterministic balanced labels; PS 94.748.
`eth_high_moment`	Independent 94.748 certificate and low-degree variants.
`eth_matching_refinement`	Matching-potential packing; a third 94.748 derivation.
`eth_packet_refinement`	Improved pair-load scale and padding; PS 95.664.
`eth_recursive_refinement`	Residue-class mode refinement; PS 133.207.
`eth_joint_cauchy`	Group-preserving full-block Cauchy modes; slope four; cached matrix option.

Packet	Contribution or limiting qualification
`sat_adaptive_contraction`	Streamed separators, PS 55.709; adaptive cached 36.750; exact kernels.
`sat_frontier_improvements`	Variable arity, PS 63.857; cached 38.485.
`subexponential_sat_frontier`	PS 63.852; cached 37.930; exact labeling and precision-peeling kernels.
`subexponential_sat_kernel_improvements`	Adaptive Smith, Gray, antipodal, and projective kernels; no smaller global order.
`subsat_contraction_tree`	Binary-built cached tables and exact ring bridge to a fast matrix backend; 35.355.

1.3 All nine incoming packets

The newer arrivals materially change the benchmark. In particular it would be incorrect to compare this iteration only against the filed 35.355 or 55.709 figures.

Archive stem	PS time	Cached time	Cached space
`3sat_round3_frontier_20261027`	63.75736	37.93011	33.62304
`ETH_Adaptive_Exact_Kernels_2026-10-27`	55.33162	35.07061	33.99950
`eth_adaptive_schedules_2026-10-26`	63.78117	37.59170	33.62304
`sat_adaptive_bounds_2026-10-26`	46.18485	34.32528	34.02418
`sat_adaptive_frontier_2026-10-26`	50.30469	31.25240	31.25240
`sat_adaptive_matrix_partition_2026-10-26`	50.21043	34.32528	34.02418
`sat_adaptive_tree_certificate_2026-10-26`	—	30.89102	30.75391
`subexponential_sat_adaptive_four_2026-10-26`	50.39373	32.91581	31.66171
`subsat_rectangular_2026-10-26`	53.97753	33.99389	33.99372

The adaptive-four packet also gives a slightly faster optional backend variant; it remains above the best incoming generic cached bound. Low-degree specializations are retained as separate branches. All nine incoming packets retain the general SAT logarithmic order. Several already analyze precision peeling, asymmetric Hermite multiplicities, conserved incidence degree, and full-table barriers. Those observations are not claimed as new here.

1.4 The SAT order that remains established

With the imported reductions, the strongest proved order in the reviewed corpus for fixed clause width is

$$(n+m)^{O_k(1)}2^{O_k\left(\frac{n}{\sqrt{\log n}}\min\{\log(2+m/n),\log\log n\}\right)}. \tag{1}$$

The direct clause-count branch can also be retained when it is smaller. At fixed density this gives exponent $O_k(n/\sqrt{\log n})$; at $m=\Theta(n\log\log n)$ it gives $O_k(n\log\log\log n/\sqrt{\log n})$; in the unrestricted worst case it gives $O_k(n\log\log n/\sqrt{\log n})$ [4, 3]. The new contraction coefficient changes constants after applying these reductions. It does not replace the square-root logarithm by a logarithm.

2 Exact interface and paid contraction costs

2.1 Imported algebra and packing

We use the same normalized problem as the current best generic packets. The input is a bipartite Boolean tensor network with at most s edges, maximum degree at most $\lceil s^{9/10} \rceil$, and entries in

$$(\mathbb{Z}/2^{P_0}\mathbb{Z})[V]/(V^{\ell}), \qquad 1 \le P_0, \ell \le s.$$

Entry evaluation has time $(s+\mathcal{L})^{O(1)}2^{O(s^{9/10})}$ and polynomial space, with fixed interface constants. The unchanged contraction proof works in a polynomial-rank extension ring with an irreducible residue polynomial of degree $O(\log s)$. Tensor descriptions retain their specified creation precision and canonical entry representatives. Lifting those representatives is not assumed to preserve ring operations.

The imported ingredients are exact Cauchy deformation and recovery, asymmetric near-unit padding, full-orientation Cauchy modes, and recursively evaluable group factors [3, 5, 6, 8]. This report alters how the finite group-factor sum is evaluated. It is not an independent proof of the entire upstream SAT algorithm.

At an active call use wire budget M, group count r, taper length u, cumulative packet exponent a, a valuation $i < u$, and $D = 2^{a-i}$. The common interface supplies

$$p_t \le (6 + o(1))MD/u, \tag{2}$$

$$M_{\text{child}} \le (2/r + o(1))M. \tag{3}$$

For each directed pair of different groups, the multiplicity of each mode index q is at most $(2+o(1))M/(r^2D)$ and its certified weight is $a + 1 - i + 4q - 2\,\mathrm{popcount}(q)$. All error terms are uniform on the fixed set of arities and the active levels of the final schedule.

If $Q = r^a$ and R, C bound the two sides' degrees, sufficient padding gates are

$$QR/M = o(1), \quad Q^2C(\log s)^2/M = o(1), \quad 2^aQ^2/M = o(1). \tag{4}$$

At the root both sides have degree at most $s^{9/10+o(1)}$. Subsequently the old side retains that bound and the fresh side has degree at most $2^{a_{d-1}+1}$. Section 4 checks these gates before using the child-size conclusion.

2.2 Exact mode lists and the entropy envelope

Enumerate exact nonnegative pair budgets b_{gh} with sum less than p_t. There are polynomially many such arrays because r is fixed and p_t is polynomial in s. Let $\mathcal{I}_{gh}$ be the finite list of mode exponent tuples of exact weight b_{gh}. Empty lists contribute zero. Put $h_{gh} = \log_2 |\mathcal{I}_{gh}|$ and

$$\kappa = \frac{\pi\sqrt{2}}{\ln 2}, \qquad \mathcal{K}(M,u) = \kappa M/\sqrt{u}.$$

The contribution of the budget array is exactly

$$\sum_{\xi_{gh} \in \mathcal{I}_{gh}} \prod_{g=1}^{r} Z_g((\xi_e)_{e \ni g}). \tag{5}$$

Each Z_g entry requires a real recursive contraction; it is not a unit-cost oracle.

For a fixed nonnegative directed weight vector z, the inherited bound is

$$\sum_e z_e h_e \le \left(\frac{\|z\|_2}{r} + o(1) \right) \mathcal{K}(M, u). \tag{6}$$

For clarity, its leading constant follows by lowering mode weights to $1 + 4q - 2\,\mathrm{popcount}(q)$ and applying the generating-function estimate

$$\sum_{q \ge 0} -\ln(1 - e^{-\theta(1+4q-2\,\mathrm{popcount}(q))}) = \frac{\pi^2}{24\theta} + O(\log^2(2/\theta)).$$

Using a common tilt θ/z_e gives leading natural-log bound

$$\theta p_t + \frac{\pi^2}{24\theta} \frac{(2 + o(1))M}{r^2 D} \sum_e z_e^2.$$

Optimize in θ, apply (2), and divide by $\ln 2$. The logarithmic guard ensures that the residual term is lower order. For a fixed finite family of vectors, the error is uniform. All adaptive arguments below use such a finite family; no pointwise bound is silently applied to an uncontrolled growing collection.

List sizes and indexing are explicit. The unbounded-knapsack recurrence

$$N(j, b) = N(j - 1, b) + N(j, b - w_j), \quad N(0, 0) = 1$$

with zero values for impossible budgets counts exact tuples. It also gives rank/unrank and polynomial-delay enumeration. Budgets, list-count bit lengths, and the number of mode types are polynomial in the original size parameter.

2.3 Matrix operations and support vectors

For disjoint group sets A, B and $O = [r] \setminus (A \cup B)$, merging two boundary tables has three independent index classes: pairs joining A to O, A to B, and B to O. Write their logarithmic dimensions as H_1, H_2, H_3. Using the established strict rational matrix exponent $\omega < 12/5$ gives the upper exponent

$$H_1 + H_2 + H_3 - \frac{3}{5} \min(H_1, H_2, H_3) = \max_j \left(\frac{2}{5} H_j + \sum_{k \ne j} H_k \right). \tag{7}$$

Square blocking proves this rectangular estimate. The cited matrix result is stronger than needed; no newest-record assertion is necessary [12].

The ring bridge is the one already established in [7, 8]. Expand each ring entry into its polynomially many integer coordinates, perform coordinate matrix products modulo sufficiently many ordinary primes away from the fixed denominator exceptions of a rational bilinear algorithm, reconstruct by CRT, apply the ring structure constants, and reduce modulo 2^{p_t}. All reconstruction bounds have polynomial bit length. This uses neither division by a nonunit in the original ring nor a false field model for that ring. The fast backend is a theoretical dependency; the small prototype in this archive uses ordinary exact multiplication.

For an undirected pair $e = \{g, h\}$ put $g_e = h_{gh} + h_{hg}$. Multiply (7) by five. Each operation supplies three vectors $v \in \{0, 2, 5\}^{\binom{r}{2}}$, zero on edges internal to its three parts, two on a discounted

cross class, and five on the other classes. A plan first divides the groups into three blocks, builds each block table by binary merges, then contracts the final triangle. If $\mathcal{V}_j$ is the complete support set for plan j, put

$$F_j(g) = \frac{1}{5} \max_{v \in \mathcal{V}_j} v \cdot g.$$

Every boundary table is the input or output of an operation. Its exponent is a sum of two of the H_i, hence at most (7). Thus F_j pays for all local time and storage after singleton caching.

Lemma 2.1 (Paid adaptive support certificate)**.** *For q fixed plans on r groups let*

$$Q_* = \max_{v_j \in \mathcal{V}_j} \left\| \sum_{j=1}^{q} v_j \right\|_2^2, \qquad \beta_r = \frac{\sqrt{2Q_*}}{5qr}, \qquad \alpha_r = \frac{\sqrt{2(r-1)}}{r}.$$

Choosing a plan with smallest F_j gives the recurrence

$$T_d \le \mathrm{poly}(s) \left[2^{(\alpha_{r_d} + o(1))\mathcal{K}_d} T_{d+1} + 2^{(\beta_{r_d} + o(1))\mathcal{K}_d} \right], \tag{8}$$

and local storage at most $\mathrm{poly}(s) 2^{(\beta_{r_d} + o(1))\mathcal{K}_d}$.

Proof. The minimum of the F_j is at most their average, which equals

$$\frac{1}{5q} \max_{v_j \in \mathcal{V}_j} \left(\sum_j v_j \right) \cdot g.$$

Give both directed orientations of an edge the weight $(\sum_j v_j)_e/(5q)$ and apply (6). Its squared directed norm is $2\| \sum_j v_j \|_2^2/(25q^2)$, giving β_r. There are finitely many support tuples. The storage assertion follows from the preceding input/output observation. Caching one group's entries involves $2(r-1)$ incident directed pairs, giving α_r; each entry is charged its child contraction. The fixed number of groups is absorbed by the polynomial factor.

Selection needs no real-number oracle. Compute $\max_{v \in \mathcal{V}_j} \prod_e (|\mathcal{I}_{gh}|\,|\mathcal{I}_{hg}|)^{v_e}$ and compare those positive integers. Their bit lengths are polynomial. Sorted sequential table scans and reshaping add only polynomial factors in index and record length, so the analysis does not rely on free exponential random access. □

3 The affine-plane contraction certificate

3.1 Four geometric plans

Identify nine group labels with $(x, y) \in \mathbb{F}_3^2$, using integer label $3x + y$. For each of the four linear forms

$$x, \qquad y, \qquad x + y, \qquad x + 2y$$

partition the points into the three level sets. Each set is a line of three points. Within an ordered line (a, b, c) first merge b, c, then merge the result with a. Finally contract the three line tables. There are six internal merges and one final triangle, hence 21 support vectors per plan.

Direction	Ordered line blocks
x	$(0,1,2) \mid (3,4,5) \mid (6,7,8)$
y	$(0,3,6) \mid (1,4,7) \mid (2,5,8)$
$x+y$	$(0,5,7) \mid (1,3,8) \mid (2,4,6)$
$x+2y$	$(0,4,8) \mid (2,3,7) \mid (1,5,6)$

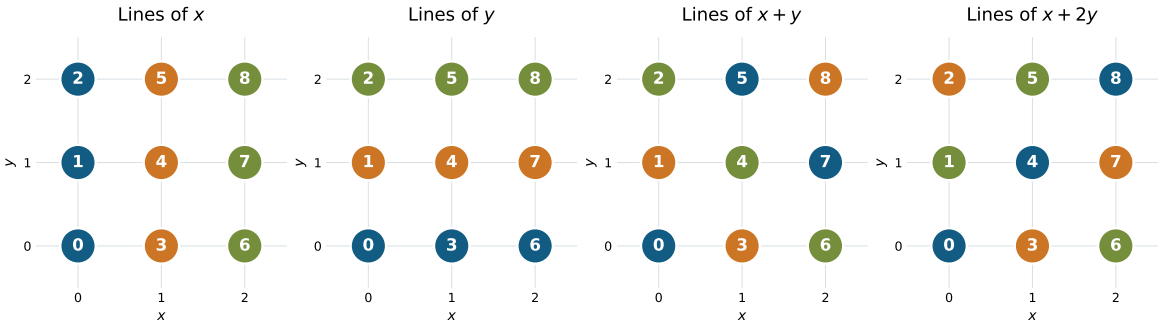


Figure 1: The same nine groups in four partitions. Equal colors form a line block. Every pair of distinct points lies together in exactly one partition, and lines from different partitions meet in one point.

These balanced intersection properties are what an arbitrary handful of permuted trees lacks. They control the overlap between *all* operation supports, including the operations used to build the tables.

3.2 A short Gram certificate

Classify an operation as P (merge two points), L (complete a line), or R (the root triangle). The following are exact maxima for these four plans. A transverse pair means supports from different parallel classes.

Type	P	L	R
Maximum single-support squared norm	354	458	486

Maximum transverse inner product	P	L	R
P	204	208	215
L	208	240	260
R	215	260	288

The single-support entries follow directly from the cross-class sizes. For example, a line-completion operation has undirected class sizes $2, 6, 12$, so its largest norm is $4 \cdot 2 + 25(6+12) = 458$. A root has three classes of nine edges, giving $9(4+25+25) = 486$. The transverse entries are exact nine-point incidence counts. The independent checker reconstructs them from Cartesian edge classes. For the theorem, a short geometric argument supplies slightly weaker mixed bounds, all still strictly below 288; no exhaustive computation is needed for its principal inequality.

A P profile is supported on edges incident to its chosen pair. An L profile is supported on edges incident to its line, except for the edge between the two points merged first. Transverse lines meet in one point. Two P supports therefore share at most nine edges, and a P and an L share at most

ten. Since coefficients are at most five, their inner products are at most 225 and 250. In a P–R comparison, the selected points lie in distinct root blocks; eleven root-crossing edges meet the pair, giving at most 275.

Two full line-incidence sets share twelve edges: eight incident with their intersection and four joining their remaining pairs. If the intersection is not the singleton in at least one of the two line merges, an omitted internal edge is shared, leaving at most eleven edges and bound 275. Otherwise the intersection is the singleton of both merges. Every possible discounted class contains at least two shared edges; each has product at most ten instead of twenty-five, so the bound is $300 - 30 = 270$.

For L–R, the transverse line meets each root block once. Root weights sum to 48 on its external incident edges and 12 on its internal triangle. The line profile omits an internal edge of root weight at least two, so its undiscounted dot product is at most $5(60 - 2) = 290$. Its discounted class has root-weight sum at least four, reducing this by at least $3 \cdot 4$ to 278. Thus every mixed transverse inner product is strictly less than 288. The root-root case has the exact geometric proof below. These arguments work for any ordering within a line.

Theorem 3.1 (Sharp four-plan certificate)**.** *For the four affine-plane plans,*

$$\max_{v_j \in \mathcal{V}_j} \|v_1 + v_2 + v_3 + v_4\|_2^2 = 5400.$$

Consequently their local coefficient in Lemma 2.1 is $\beta_9 = 1/\sqrt{3}$.

Proof. The Gram bounds give

$$\left\| \sum_j v_j \right\|_2^2 = \sum_j \|v_j\|_2^2 + 2 \sum_{j<k} \langle v_j, v_k \rangle \le 4 \cdot 486 + 2 \binom{4}{2} 288 = 5400.$$

Every choice of four root supports attains equality. Here is a geometric verification, also explaining why the constant is simple. Let C_j be the cross-edge set of direction j and let D_j join the two parallel lines chosen for its discounted class. A root support is $5\mathbf{1}_{C_j} - 3\mathbf{1}_{D_j}$. For distinct directions,

$$|C_j \cap C_k| = 18, \quad |D_j \cap C_k| = |C_j \cap D_k| = 6, \quad |D_j \cap D_k| = 2.$$

The last identity counts the two diagonals of the affine 2 by 2 rectangle obtained by retaining two levels of each independent linear form. Therefore every transverse root inner product is $25 \cdot 18 - 15(6 + 6) + 9 \cdot 2 = 288$. Each root norm is 486. Finally

$$\beta_9^2 = \frac{2 \cdot 5400}{25 \cdot 4^2 \cdot 9^2} = \frac{1}{3}.$$

□

As an additional independent check, all $21^4 = 194{,}481$ tuples were enumerated with integer arithmetic. The sum of their squared norms is 622,974,240. Exactly $81 = 3^4$ tuples attain 5400; every one uses four root supports. The largest value involving an internal operation is 5204. These checks support the finite incidence table; they are not samples of entropy vectors.

3.3 Sharpness within this particular menu

The coefficient is the exact minimax value for this four-plan support model, not merely a convenient sufficient bound. Choose any root support v_j from each direction and set $g = \sum_j v_j$. Then $\|g\|_2^2 = 5400$ and

$$v_j \cdot g = 486 + 3 \cdot 288 = 1350 \qquad (1 \le j \le 4).$$

Thus every plan has $F_j(g) \ge 270$. The averaging bound gives equality. Assigning each directed orientation half the corresponding g_e and normalizing to the Euclidean entropy ball witnesses the coefficient $1/\sqrt{3}$. This is not an optimality claim over all possible tree menus, over actual realizable mode lists, or over SAT algorithms.

3.4 A second finite certificate for the recursive bottleneck

For thirteen groups use the following four plans. In each block, recursively split the ordered list at its floor midpoint, build the two subblocks, and merge. Finish with the three-block triangle.

Plan	Ordered blocks
1	$(0,1,2,3,4) \mid (5,6,7,8) \mid (9,10,11,12)$
2	$(12,10,1,6,4) \mid (9,0,3,8) \mid (5,2,7,11)$
3	$(0,8,11,12,1) \mid (7,10,4,3) \mid (5,2,6,9)$
4	$(1,7,0,6,9) \mid (5,10,11,3) \mid (12,8,4,2)$

The first three plans occur in the incoming certificate; adding the fourth and certifying the complete new menu is the additional finite refinement. There are eleven operations and 33 profiles per plan. Direct exact evaluation gives squared norm at most 1064 for every single profile and inner product at most 647 for every pair from different plans. Consequently

$$Q_{13} \le 4 \cdot 1064 + 12 \cdot 647 = 12020, \qquad \beta_{13}^2 = \frac{2 \cdot 12020}{25 \cdot 16 \cdot 169} = \frac{601}{1690}.$$

Equality occurs when each final triangle discounts the class between its two four-point blocks. The $6534 = 6 \cdot 33^2$ pairwise integer checks and the matching witness suffice to certify this bound. Two independently written programs also enumerate all $33^4 = 1{,}185{,}921$ tuples; their squared-norm sum is 5,700,575,232, with a unique maximizer. No floating search result is used as a mathematical acceptance condition.

For completeness, the finite Gram calculation is exactly the following:

```
for plan in plans:
    operations = binary_merges_inside_blocks(plan) + [final_triangle(plan)]
    profiles = [cross_class_weights(op, discount, high=5, low=2)
                for op in operations for discount in range(3)]
check(all(dot(v,v) <= 1064 for v in every_profile))
check(all(dot(v,w) <= 647 for v,w in different_plan_pairs))
check(dot(sum_of_four_displayed_root_profiles,
          sum_of_four_displayed_root_profiles) == 12020)
```

The supplied plans, midpoint rule, pair coordinates, and profile definition fully specify this finite certificate.

3.5 Why a better local bound is not enough

The incoming best cached schedule has its maximum at the second recursion stage, not at the root. Replacing its nine-group coefficient alone would improve memory but leave the time coefficient $30.8910189\ldots$ unchanged. The global construction must also improve that later bottleneck. The next section includes the needed second certificate, its actual child-entry costs, and an explicit feasible schedule. This prevents a local-only gain from being reported as a global speedup.

4 A fully paid recursion with the new certificates

The local affine-plane certificate becomes a global improvement only after the expensive evaluation of the cached group functions is included. We use the explicit $9 \to 13 \to 7$ schedule below to improve the second-stage bottleneck as well as the root. This section states all additional resource assumptions and checks the complete recurrence.

4.1 Normalized inherited interface

Let $s \geq 2$, $\lambda = \lceil \log_2 s \rceil$, and let the initial bipartite Boolean tensor network have at most s wires and maximum degree at most $\lceil s^{9/10} \rceil$. Its entries lie in $(\mathbb{Z}/2^{P_0}\mathbb{Z})[V]/(V^{\ell})$ with $P_0, \ell \leq s$. Let $\mathcal{L}$ be its description length. The permitted original entry procedures take $(s+\mathcal{L})^{O(1)}2^{O(s^{9/10})}$ time and polynomial space, with fixed exponents and constants. The ring extension, canonical creation-time lifts, exact deformation, and Hermite recovery are the inherited normalized interface of the reviewed packets. Thus the result below is a proved refinement of that interface; it does not independently verify the upstream exact-counting algorithm.

At an active call with M wires, group count r, and taper length u, write

$$\mathcal{K}(M,u) = \kappa \frac{M}{\sqrt{u}}, \qquad \kappa = \frac{\pi\sqrt{2}}{\ln 2}.$$

The inherited full-Cauchy entropy estimate says that if h_{gh} is the base-two logarithm of the exact list cardinality on directed group pair (g,h), then for every fixed nonnegative weight vector z,

$$\sum_{g \neq h} z_{gh} h_{gh} \leq \left(\frac{\sqrt{\sum_{g \neq h} z_{gh}^2}}{r} + o(1) \right) \mathcal{K}(M,u).$$

Exact weight allocations have total below the polynomially bounded working precision. Because only the three fixed group counts $9, 13, 7$ occur, there are polynomially many such allocations. The local certificates from the preceding section apply uniformly to each allocation.

There are $2(r-1)$ directed pairs incident to one group. Consequently the total number of singleton-table entries, including the sum over all groups, is at most

$$\mathrm{poly}(s)\, 2^{(\alpha_r + o(1))\mathcal{K}(M,u)}, \qquad \alpha_r = \frac{\sqrt{2(r-1)}}{r}.$$

Every entry is computed by a recursive child contraction with at most $(2/r + o(1))M$ wires. Charging these calls separately from the local tree operations gives

$$T_d \leq \mathrm{poly}(s) \left[2^{(\alpha_{r_d} + o(1))\mathcal{K}_d} T_{d+1} + 2^{(b_{r_d} + o(1))\mathcal{K}_d} \right], \qquad \mathcal{K}_d = \mathcal{K}(M_d, u_d). \tag{9}$$

The local coefficients used here are

$$b_9^2 = \frac{1}{3}, \qquad b_{13}^2 = \frac{601}{1690}, \qquad b_7^2 = \frac{632}{1225}.$$

The first two come from the new four-plan certificates. The last is the fixed seven-group plan $(0,1,2) \mid (3,4) \mid (5,6)$: its integer operation supports have maximum squared norm 316, so its directed normalized coefficient is $2 \cdot 316/(25 \cdot 7^2)$. The supplied checker verifies this bound directly.

The matrix operations use the inherited exact integer-coordinate and CRT bridge for a fixed rational matrix multiplication algorithm with exponent strictly below 12/5. Thus the proof pays for the arithmetic over the source ring rather than assuming that the ring is a field. No fast matrix multiplication backend or end-to-end contraction solver is implemented by the certificate programs.

4.2 An explicit strict rational schedule

Choose

$$\begin{aligned} r_0 &= 9, & \eta_0 &= \frac{15773}{1000000}, \\ r_1 &= 13, & \eta_1 &= \frac{9}{800}, \\ r_d &= 7 \quad (d \geq 2), & \eta_d &= \frac{8597}{1800000}\left(\frac{4}{9}\right)^{d-2} \quad (d \geq 2). \end{aligned}$$

Put

$$u_d = \lfloor \eta_d \lambda \rfloor, \qquad B = \lceil 6 \log_2 \lambda \rceil, \qquad a_d = \sum_{j=0}^{d} (u_j + B), \qquad H = \left\lceil \frac{s}{\lambda^{3/4}} \right\rceil.$$

Use the inherited near-unit padding with relative slack λ^{-3}. Calls with $M \leq H$ are evaluated directly. For sufficiently large s every active u_d is positive; a fixed exceptional initial range is handled by exhaustive contraction. The exact capacity sums are

$$\eta_0 + \eta_1 = \frac{27023}{1000000}, \qquad \sum_{d \geq 0} \eta_d = \frac{1781}{50000}.$$

The following integer inequalities give rational upper bounds for every logarithm needed in the packing proof:

$$9^{100000} < 2^{316993}, \qquad 13^{100000} < 2^{370044}, \qquad 7^{100000} < 2^{280736}.$$

They are checked using exact arbitrary-precision integers. Their respective upper bounds for $\log_2 9$, $\log_2 13$, and $\log_2 7$ are 316993/100000, 370044/100000, and 280736/100000.

At the root both old sides have degree exponent 9/10. At a later depth, one side has that same old-degree bound, whereas the fresh side has degree at most $2^{a_{d-1}+1}$. The asymmetric packing requirements are, up to polylogarithmic factors, $Q^2\Delta$ at the root and QR, Q^2C, 2^aQ^2 later, with $Q = r^a$, retained degree $R \leq s^{9/10+o(1)}$, and fresh degree $C \leq 2^{a_{d-1}+1}$. The dominant margins are

Stage	Upper bound on the leading exponent	Margin below one
Root	$\frac{9}{10} + 2\frac{15773}{1000000}\frac{316993}{100000}$	$\frac{69411}{50000000000}$
Depth one	$\frac{9}{10} + \frac{27023}{1000000}\frac{370044}{100000}$	$\frac{75247}{25000000000}$
Tail	$\frac{9}{10} + \frac{1781}{50000}\frac{280736}{100000}$	$\frac{287}{156250000}$

Every margin is strictly positive. The secondary fresh-side and packet terms are controlled by

$$\frac{1781}{50000}\left(2\frac{370044}{100000}+1\right)<1.$$

The fresh packet degree exponent is at most $1781/50000+o(1)<9/10$. The smallest displayed power gaps are of order 10^{-6}. They certify eventual asymptotic feasibility, with no useful practical-size threshold claimed here. This is one reason the coefficient comparison should not be interpreted as a benchmark prediction.

4.3 Uniformity, depth, and precision

We give a bootstrap rather than using a depth bound as an unproved premise of packing. Consider hypothetical active tail depths up to $3\log_3\lambda/4+O(1)$. On this range,

$$u_d\geq\lambda^{\epsilon+o(1)},\qquad \epsilon=1-\frac{3\ln(9/4)}{4\ln 3}>0.$$

For example, positivity follows from $(9/4)^3<3^4$. The cumulative guard adds only $O((\log\lambda)^2)$ to a_d. The positive fixed margins above absorb this guard, all packing polylogarithms, and $s/M<\lambda^{3/4}$. Padding is therefore valid throughout the hypothetical range, with relative error $O(\lambda^{-3})+s^{-c+o(1)}$ for a fixed $c>0$. In the tail every child has at most $(2/7+o(1))M<M/3$ wires eventually. Hence an active tail cannot extend beyond the stated hypothetical range. This closes the bootstrap and proves the actual depth is $O(\log\lambda)$.

All active taper lengths tend to infinity uniformly. Floors therefore change their leading terms by $o(1)$ relatively. The accumulated padding error over the path is $1+o(1)$, and consequently

$$M_0\leq s,\qquad M_1\leq(1+o(1))\frac{2s}{9},\qquad M_d\leq(1+o(1))\frac{4s}{117}\left(\frac{2}{7}\right)^{d-2}\quad(d\geq 2).\tag{10}$$

For a retained valuation $i<u_d$, set $D=2^{a_d-i}$. Then $D\geq 2^{a_{d-1}+B+1}$. The inherited input precision of the call is at most $O(s\lambda^2 2^{a_{d-1}})$. Relative to MD/u_d, it is bounded by

$$O\Big(\lambda^{15/4}2^{-B}\Big)=o(1).$$

The ordinary Hermite evaluation loss is lower order as well. The inherited sharp precision bound $p_t\leq(6+o(1))MD/u_d$, and hence the weighted entropy estimate, therefore hold uniformly over all active calls. All precisions retain polynomial bit length, since $a_d\leq(1781/50000)\lambda+O((\log\lambda)^2)$.

Direct leaf contraction costs $2^{O(s/\lambda^{3/4})}$ times the entry cost. Original entries cost $2^{O(s^{9/10})}$ times a fixed polynomial. Fresh packet entries cost $\exp(O(2^{a_d}\log s))=\exp(s^{1781/50000+o(1)})$. These terms are all $2^{o(s/\sqrt{\lambda})}$. Their immutable creation-time descriptions are retained; later entry evaluation does not recursively expand earlier contraction routines. The ancestor cache costs multiplying the leaf work remain explicitly charged in (9).

4.4 Exact evaluation of the recurrence

Define the leading coefficients of $\mathcal{K}_d$ on the common scale $s/\sqrt{\lambda}$ by

$$K_0^*=\frac{\kappa}{\sqrt{\eta_0}},\qquad K_1^*=\frac{2\kappa}{9\sqrt{\eta_1}},\qquad K_2^*=\frac{4\kappa}{117\sqrt{\eta_2}}.$$

After depth two the ratio is

$$\gamma = \frac{2/7}{\sqrt{4/9}} = \frac{3}{7}.$$

Unrolling the paid recurrence gives the maximum of three quantities:

$$\begin{aligned} C_0 &= b_9 K_0^*, \\ C_1 &= \alpha_9 K_0^* + b_{13} K_1^*, \\ C_\infty &= \alpha_9 K_0^* + \alpha_{13} K_1^* + K_2^* \max\left\{ b_7, \frac{\alpha_7}{1-\gamma} \right\}. \end{aligned}$$

Indeed, the finite tail aggregation branches have tail part

$$K_2^* \left(\alpha_7 \frac{1-\gamma^j}{1-\gamma} + b_7 \gamma^j \right), \qquad j \geq 0,$$

and the terminal leaf route is the limit. Here $\alpha_7/(1-\gamma) = \sqrt{3}/2 > b_7$, so the limit controls every tail branch. This also pays for the route containing all recursive cache calls.

Exact radical expressions for these three coefficients are

$$\begin{aligned} C_0 &= \frac{\pi}{\ln 2} \sqrt{\frac{2000000}{47319}}, \\ C_1 &= \frac{\pi}{\ln 2} \left(\sqrt{\frac{32000000}{1277613}} + \sqrt{\frac{384640}{123201}} \right), \\ C_\infty &= \frac{\pi}{\ln 2} \left(\sqrt{\frac{32000000}{1277613}} + \sqrt{\frac{51200}{41067}} + \sqrt{\frac{1600000}{4358679}} \right). \end{aligned}$$

The supplied exact-rational interval calculation gives

Branch	Coefficient
C_0	$29.46603025659479065850\ldots$
C_1	$30.69133782428594886802\ldots$
C_∞	$30.48972822403429948939\ldots$

In particular C_1 is the unique maximum of the three. The computation encloses π by Machin's identity and alternating arctangent series, $\ln 2$ by the positive series $2\,\mathrm{artanh}(1/3)$ with an explicit tail bound, and square roots by integer square-root enclosures. The interval endpoints are outward rounded; the table displays approximations. Floating-point approximations are not used in any acceptance check.

For space, local tables are bounded by the chosen tree's local operation bound, including singleton caches. Ancestor tables stored on one recursive path add in size; their exponents do not add. Thus the space coefficient is

$$\max\{b_9 K_0^*,\ b_{13} K_1^*,\ b_7 K_2^*\} = C_0,$$

since the latter two quantities are respectively $8.0083771568264\ldots$ and $2.2775442472698\ldots$. The fixed polynomial factors across $O(\log \lambda)$ levels contribute only $O(\lambda \log \lambda) = o(s/\sqrt{\lambda})$ to the exponent. The dependence on the original description length remains one fixed polynomial, because branching is independent of $\mathcal{L}$.

Theorem 4.1 (New generic normalized certificate)**.** *Under the inherited normalized interface specified above, the affine-plane and four-plan construction contracts the root network deterministically in*

$$(s+\mathcal{L})^{O(1)}2^{(30.691337824285948868 02...+o(1))s/\sqrt{\lambda}}$$

time and

$$(s+\mathcal{L})^{O(1)}2^{(29.466030256594790658 50...+o(1))s/\sqrt{\lambda}}$$

space. The eventual coefficients 30.692 *and* 29.467*, respectively, are valid. The method uses subexponential space.*

Proof. The local selectors merely reassociate exact finite sums and products in the inherited mode-contraction identity. Every feasible exact-weight allocation is evaluated once. The strict packing, depth, precision, and entry-access checks above justify all inherited calls. The paid recurrence and local storage calculation give the asserted resources, including leaves and polynomial overhead. □

Compared with the reviewed generic cached certificate $30.891018920366\ldots$ in time and $30.753912832055\ldots$ in space, both leading coefficients decrease. This is a comparison of explicit upper bounds per normalized wire; it is neither a measured SAT speedup nor a coefficient per SAT variable. The logarithmic order $s/\sqrt{\log s}$, and consequently the inherited unrestricted SAT order, remain unchanged. No global optimality of the finite portfolios or schedule is claimed.

5 The supplied Anthropic formalization changes the evidence

During this investigation the user supplied pull request 31 of `anthropics/formal-math`, specifically commit `1a6eee293b9e00cb432eef6018a21f523623a9af`. This is substantive new evidence, not merely another statement of an unpublished proof. The ProveIt README at the initial review pin said that the Claude manuscript's Lean development was not public. The closing commit has now corrected that statement and linked this public development [13, 2].

5.1 What the public theorem actually states

`eth/MainResults.lean` defines the result over Mathlib's finite-alphabet multistack Turing machines. It quantifies one machine and one constant before the clause width and the formula. For $k \geq 1$, $n \geq 4$, and at least two clauses, under the stated encoding and formula-normalization conditions, the machine outputs the exact number of satisfying assignments within

$$\left\lfloor 2^{C\min\{km/\sqrt{\log_2 m},\, k^2 n \log_2\log_2 n/\sqrt{\log_2 n}\}} \right\rfloor$$

steps [14]. Every declared variable must occur, clauses have width at most k, and identical clause lists may not be repeated. Clauses can contain repeated literals, opposing literals, or be empty. Small exceptional sizes, unused variables, and duplicate clauses need ordinary preprocessing or direct treatment before applying the formal theorem. The input encoding uses unary width/variable counts and fixed-width literal slots; the output uses $n+1$ binary positions.

`eth/Solution.lean` proves both `MainResults.mainTheorem` and `MainResults.ethIsFalse` without hypotheses. The first proof follows the counted route through the implementation-level theorem; the second uses the normalization and Turing-machine consequences. The intermediate cost model permits polynomial-time whole-state work at an abstract step, but the final proposition

counts actual Mathlib machine steps and absorbs the simulation cost into the unspecified constant [15]. No polynomial-space guarantee is part of this displayed main proposition. Its unknown constant therefore cannot be equated with one of the normalized-wire coefficients in this report.

5.2 Verification evidence and its limits

The public source tree contains 1052 Lean files: 1050 development modules and two axiom-printing helper files. The source-only import audit found 810 local modules reachable from `Solution.lean`; this closure excludes `Challenge.lean`. The latter deliberately supplies two `sorry` targets for Comparator. They are not proof holes imported by the solution.

More strongly, the published `verify (eth)` job succeeded. Its retrieved log records a successful Lake build, Comparator acceptance, and nanoda kernel acceptance. The checked merge commit has Git tree `ab1069cc17ae5088e175ec37fa16fe8bd273753d`, identical to the tree of the user-supplied commit. The evidence therefore concerns exactly the supplied source tree, not a nearby earlier version [16]. The comparator configuration names the two target theorems and allows only the standard axioms `propext`, `Quot.sound`, and `Classical.choice`.

We retrieved and inspected this source and the published verification evidence. We did not perform a fresh local build of that development. The new affine-tree lemmas and improved numerical schedule in this report are separate natural-language results with exact computational certificates; they are not part of that Lean verification.

5.3 Why this does not automatically give a better order

The Anthropic manuscript supplies another contraction route, using crossing replacements and graded partition spaces. Its displayed worst-case counting order agrees with the square-root-logarithm scale being investigated here. The paper itself identifies a possible structural lever: a replacement with graded entropy $\log \dim V_E = O(E^\alpha)$, $\alpha < 1/2$, together with all required effective identities and factorizations, would improve the logarithmic denominator power. Constructing only a smaller graded space does not supply those interfaces.

The existing second-round corpus already discusses this route and its characteristic-zero Heisenberg and locality obstructions. Those observations must not be presented as new discoveries caused by seeing the now-public Lean code. The new source is valuable because it exposes verified interfaces against which a genuinely new replacement could be tested.

A tempting composition also needs careful scale accounting. Suppose a growing-arity Cauchy reduction has entropy scale $B = O(M\sqrt{\log r/\log s})$, while a clique engine saves a factor $(\log r)^\alpha$. Their direct composition has upper exponent

$$O\left(\frac{M(\log r)^{1/2-\alpha}}{\sqrt{\log s}}\right).$$

At $\alpha = 1/2$ these two logarithmic factors cancel. Even this calculation assumes valid oracle, precision, and nonuniform-list interfaces. It explains why inserting the available clique solver is not by itself a proof of a new logarithmic order; it is not a lower bound against every possible composition.

6 What was tried to obtain a smaller exponent order

The objective of the investigation was stronger than another small coefficient improvement. Several routes were examined specifically because they might remove an entire logarithmic factor. Their status is recorded here to distinguish a missing proof from a completed improvement.

6.1 Lower initial precision and conserved incidence degree

The compressed dense reduction has wire budget $M = O(n \log\log n)$ but initial counting precision only $n + 1$. This suggests exploiting $P_0 \ll M$. However the generic deformation has degree $M(2L - 1)$; Hermite recovery introduces precision of leading size $6MD/u$ regardless of that smaller initial precision. Merely replacing P_0 by $n + 1$ does not change the dominant term.

There is a genuine algebraic alternative: if a degree bound $K = O(n)$ were retained through the compressed deformation and every recursive entry interface, the local mode entropy could scale as $\sqrt{MK/u}$ instead of $M/\sqrt{u}$. With $M = O(n \log\log n)$ this would target $n\sqrt{\log\log n/\log n}$. The reviewed corpus already states this conditional route. Its recent degree-flow lemma shows that incoming mode extraction removes exactly the degree allowance transferred to outgoing linear forms. That conservation identity is useful but does not close the algorithmic representation.

Binary ranks of low-weight subsets do not automatically give low Boolean polynomial degree. Carrying the original incidence polynomial preserves degree but loses the compressed wire count; carrying a whole grouped coefficient as one implicit tensor can make entry extraction expensive. For example, a coefficient of a product of linear forms can be a permanent. That example is an interface warning, not a hardness proof for the actual Cauchy tensors. We did not construct the missing uniform compiler.

6.2 Precision peeling and ordinary tree improvements

An exact mode contribution with total valuation V needs only residual precision $p_t - V$. This is already an effective kernel technique in the corpus. It does not alone justify restarting every interpolation scale. Budget layers with $p_t - V = p_t/(\log s)^B$, for fixed B, can retain the leading weighted-partition entropy while losing only $O(\log\log s)$ in the binary logarithm of the precision. Resetting the cumulative packet exponent by $\Theta(\log s)$ would require a much larger reduction. This is a limitation of this budget-envelope argument, not a universal lower bound on precision-aware algorithms.

The new affine-plane construction also lives inside a model with a known order obstruction. Full boundary tables on a complete mode graph must include a substantial cut in the Euclidean entropy relaxation. Root packing simultaneously limits u_0 to $O(\log s/\log r)$. These statements, already proved in the incoming packets, prevent tree selection and a faster matrix exponent alone from proving $o(s/\sqrt{\log s})$ in that certificate model [10, 8]. Our improvement is consistent with that obstruction: it decreases the coefficient while retaining its order.

6.3 A new scoped obstruction for direct odd-prime replacement

Changing from two-adic to p-adic arithmetic initially looks attractive: an exact count needs fewer base-p digits, and a growing characteristic might appear to save a factor of $\sqrt{\log p}$. The binary locality mechanism explains why that inference is not valid for a direct replacement.

Lemma 6.1 (Leading-digit locality)**.** *Let p be prime and let $A, B : \mathbb{F}_p \to \mathcal{C}$ be row and column labels. If*

$$A(x) = B(x + c) \quad \text{for all } x \in \mathbb{F}_p, c \in C_0,$$

and C_0 contains two distinct elements, then both labels are constant.

Proof. For $c \neq d$ in C_0 one has $B(x + c) = B(x + d)$ for every x. Thus $c - d$ is a nonzero additive period of B. A nonzero element generates the additive group of the prime field, so B is constant; the displayed identity makes A constant as well. □

In the binary construction every integer of valuation i flips bit i, and there is just one nonzero leading residue. For odd p, asking one fixed nontrivial digit grouping to localize every nonzero leading residue falls under Lemma 6.1. Node-dependent groupings and other deformations are outside this statement.

One can try to retain only leading digit 1. The following exact calculation shows the resulting precision penalty for the ordinary linear taper. Work on the complete grid $\{0, \dots, p^{a+1} - 1\}$, excluding zero from the interpolation nodes. Retain a node q of valuation $j < u$ only when its first nonzero base-p digit is 1, and give it multiplicity $m_j = h(u - j)$; all other multiplicities are zero. For an active node t of valuation i set $D = p^{a-i}$ and $H = u - i$.

Proposition 6.2 (Loss of tapered cancellation)**.** *For the Hermite prefactor*

$$E_t(0) = \prod_{q \neq 0,t} \left(\frac{-q}{t - q} \right)^{m(q)},$$

its exact valuation is

$$\begin{aligned} v_p(E_t(0)) &= hD \sum_{j=1}^{a-i} \frac{j}{p^j} \big((H - j)_+ - (p - 1)H \big) \\ &= -hD \left((p - 2)H \sum_{j=1}^{a-i} \frac{j}{p^j} + \sum_{j=1}^{a-i} \frac{j \min(j, H)}{p^j} \right). \end{aligned} \tag{11}$$

For fixed odd p, at $i = 0$ this has a term of order hDu; the binary case cancels that term exactly.

Proof. Nodes of valuation less than i have equal valuations in numerator and denominator and contribute zero. For $j \geq 1$, the grid contains D/p^j nodes of valuation $i + j$ with selected leading digit; their multiplicity is $h(H - j)_+$. It contains $(p - 1)D/p^j$ nodes with $v_p(t - q) = i + j$; they all have valuation i and the same leading digit as t, hence multiplicity hH. Subtract these contributions and sum the valuation level j. The second equality uses $(H - j)_+ = H - \min(j, H)$. □

Covering a deformation of degree $\Theta(Mp^{a+1})$ requires $h = \Theta_p(M/u)$, since $\sum_{j<u} p^{a-j} h(u - j) = \Theta_p(hp^a u)$. Thus for fixed odd p the new term in (11) has size $\Theta_p(MD)$, losing the helpful $1/u$ factor. This calculation does not rule out all nonbinary approaches. It rules out the tempting argument that changing the radix, while keeping this locality and one-digit taper, automatically improves the asymptotic order.

6.4 Unbalanced binary trees do not evade the flag witness at the used arities

The incoming flag construction proves optimality among balanced binary separator trees in its Euclidean relaxation. We checked the more general tree shapes at the arities $4, 8, 16$ used by its best polynomial-space schedule. Give directed pairs incident with a distinguished vertex weight 1 and all other directed pairs weight $2/3$. Let $P(n)$ be the smallest possible maximum root-to-leaf cut cost on n ordinary vertices, and $S(n)$ the corresponding value with one distinguished vertex. Exact recurrences are

$$\begin{aligned} P(1) &= S(1) = 0, \\ P(n) &= \min_{1\le a<n} \left\{ \frac{4}{3}a(n-a) + \max(P(a), P(n-a)) \right\}, \\ S(n) &= \min_{1\le a<n} \left\{ \frac{2}{3}(n-a)(2a+1) + \max(S(a), P(n-a)) \right\}. \end{aligned}$$

In the last recurrence the distinguished child has a vertices. All labels of the same type are interchangeable, so every binary tree shape is covered; the recurrences are proved by its first split. Exact rational evaluation gives

$$S(4) = 26/3, \qquad S(8) = 98/3, \qquad S(16) = 370/3.$$

These equal the squared witness norms in the existing flag certificate. After normalization, the optimal local coefficients at these arities remain $r^{-1}\sqrt{(r-1)(4r+10)/9}$ even when unbalanced trees are allowed. This is a finite model-specific strengthening. It neither proves the same formula for every r nor lower-bounds arbitrary contraction algorithms; indeed a naive extension of the formula to every integer arity is false.

7 Verification, artifacts, and next research steps

7.1 What was actually verified

The archive contains standard-library Python checks for the new support certificates and schedule, an independent reconstruction of the affine certificate, and exact tests of the finite contraction identity. The geometric proof of Theorem 3.1 is supplemented by 2646 transverse support-pair checks and all 194,481 four-support tuples. The second tree certificate is checked exhaustively with integer arithmetic. The rational schedule checks its capacity inequalities and evaluates its coefficient expressions at high precision. Exact inequalities are used for the logarithmic packing gates, rather than assuming that a floating-point optimizer found an interior point.

The local contraction prototype evaluates the same small factor graph using every affine plan and using automatic integer-score selection. Across 24 seeded fixtures, all 96 planned contractions and 24 selected contractions agree with direct enumeration over a total of 18,029 edge assignments. Fixtures include asymmetric domains, zero factors, empty domains, signed representatives, and several moduli 2^p. Seven invalid-input controls are also checked. These are exact equality checks, not timing claims. The prototype uses ordinary arithmetic and does not implement the imported fast matrix-multiplication backend, the full Cauchy engine, or an end-to-end SAT solver.

The odd-prime prefactor identity and unbalanced-tree recurrences have independent finite checks. Checks explicitly raise errors instead of relying on Python assertions that vanish under optimization. Key outputs are reproduced with `python -O`. The final PDF is compiled from the included source and rendered for visual inspection.

7.2 Reproduction and integration

Run `python3 code/reproduce.py` from the archive root to regenerate the arithmetic checks and kernel results. No network access or downloaded repository is needed for those checks. Run `make pdf` to compile the report with a standard TeX installation. The vector illustration can be regenerated with the optional plotting script; it is already included and is not needed by the mathematical checkers.

The archive keeps the new implementation independent of copied prior scripts. The provenance directory records the reviewed pins, archive hashes, source paths, and the additional public formalization evidence. External manuscripts and the large Lean development are referenced by immutable URL rather than duplicated. A claim ledger distinguishes imported interfaces, new proved lemmas, finite checks, and unresolved targets.

For repository integration, add this packet as one research directory and record both the time and space coefficients. Preserve the now-corrected public Anthropic link and, where useful, record the matching-tree CI evidence. Retain the best polynomial-space and low-degree branches. Do not convert the new wire coefficient directly into a coefficient in n, describe the prototype as a SAT solver, or mark the new refinements Lean verified.

7.3 Further research questions

1. **Can a closed representation avoid full boundary tables?** A useful candidate must specify its construction cost, entry and summation algorithms, immutable precision semantics, and behavior under another contraction. A dimension count alone does not provide these interfaces.

2. **Can finite geometry improve the entire cost profile?** The four parallel classes give a sharp local constant for their own menu. Other resolvable designs may jointly improve root and recursive caching costs. The objective is the maximum of the fully unrolled paid recurrence, not the best isolated root coefficient.

3. **Can the geometric lemma be formalized compactly?** The nine-point incidence argument is a small, well-defined target for Lean: formalize the three operation types, their transverse overlaps, the Gram inequality, and the exact selection argument. The larger inherited contraction interface is a separate formalization task.

4. **Can actual mode-list constraints beat the Euclidean envelope?** The sharpness witness in Section 3 belongs to a relaxed entropy ball. Proving it unrealizable for recursively produced tensors, with a quantitatively stronger replacement envelope, could improve the analysis in a way ordinary tree optimization cannot.

5. **Can a different deformation evade the digit obstruction?** Lemma 6.1 leaves open node-dependent labels, additional packet structure, and nonprime residue-field constructions. A viable design must restore both nontrivial locality and the tapered $1/u$ precision gain.

6. **Can the public graded-space interface be strengthened?** The Anthropic development offers precise effective contracts. A new graded construction must improve its entropy while preserving the crossing identities, both factorizations, coefficient bit lengths, and charged evaluation costs. Repeating the already stated entropy hypothesis would not solve this problem.

The positive result of this iteration is a certified, geometrically motivated improvement of the explicit cached upper bound. The stronger requested breakthrough in logarithmic order remains an open objective.

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[4] *SAT density improvement* and *ETH density refinement*, first-round research packets in `Computability/SubexponentialSAT/Research/`, at the ProveIt pin in [1]. Sources `sat_density_improvement/density_aware_sat_improvement.tex` and `eth_density_refinement/report.tex`.

[5] *ETH asymmetric packet balancing*, second-round packet, `Research/eth_asymmetric_packet_balancing/report.tex`, same pin.

[6] *Full crossing-block Cauchy modes*, `Research/eth_joint_cauchy/report.tex`, same pin.

[7] *Subexponential SAT contraction tree*, `Research/subsat_contraction_tree/report/report.tex` and `report/cached_tree.tex`, same pin. Cached-table construction and exact ring bridge to a rational fast matrix algorithm.

[8] *Adaptive Contraction Trees with Exact Support Certificates*, incoming archive `sat_adaptive_tree_certificate_2026-10-26.zip`, same pin. https://github.com/VladimirReshetnikov/ProveIt/blob/73dfb5e5ab6e91bd7375e5b97ef85d7e1cc8e81d/docs/incoming/sat_adaptive_tree_certificate_2026-10-26.zip.

[9] *SAT adaptive bounds*, incoming archive `sat_adaptive_bounds_2026-10-26.zip`, same pin, especially `report/flags.tex` and `report/interface.tex`.

[10] *SubSAT rectangular*, incoming archive `subsat_rectangular_2026-10-26.zip`, same pin, especially `sections/hermite.tex`, `sections/inventory.tex`, and `sections/table_barrier.tex`.

[11] *SAT adaptive frontier*, incoming archive `sat_adaptive_frontier_2026-10-26.zip`, same pin, `sections/degree_flow.tex`; and the earlier `Research/3sat_asymptotic_analysis/research_runtime/CONDITIONAL_DEGREE_KERNEL.tex`.

[12] J. Alman, R. Duan, V. Vassilevska Williams, Y. Xu, Z. Xu, and R. Zhou, *More Asymmetry Yields Faster Matrix Multiplication*, arXiv:2404.16349v2, 2024. https://arxiv.org/html/2404.16349v2. Only the strict rational-field inequality $\omega < 12/5$ is used.

[13] Anthropic, *A $2^{o(n)}$ Time Algorithm for 3-SAT*, public paper and Lean project, user-supplied pull request 31, commit `1a6eee293b9e00cb432eef6018a21f523623a9af`. https://github.com/anthropics/formal-math/pull/31/changes/1a6eee293b9e00cb432eef6018a21f523623a9af.

[14] Anthropic, `eth/MainResults.lean`, same commit. https://github.com/anthropics/formal-math/blob/1a6eee293b9e00cb432eef6018a21f523623a9af/eth/MainResults.lean.

[15] Anthropic, `eth/Solution.lean` and `eth/proof/PaperIndex.lean`, same commit. https://github.com/anthropics/formal-math/blob/1a6eee293b9e00cb432eef6018a21f523623a9af/eth/Solution.lean.

[16] Anthropic, successful `verify (eth)` CI job, workflow run `40440389861`, job `121979112896`. The checked merge has the same complete Git tree as the supplied commit. https://github.com/anthropics/formal-math/actions/runs/40440389861/job/121979112896.